

HEWANORRA INTL, Saint Lucia

WMO: 789480

Lat: 13.7369N

Lon: 60.9515W

Elev: 10

StdP: 101.20

Time Zone: -4.00 (W04)

Period: 94-19

WBAN: 00406

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	22.4	23.1	17.9	12.9	25.4	18.8	13.6	25.7	13.2	26.5	12.4	26.5	3.2	40	0.675

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	5.2	31.8	26.6	31.2	26.4	31.1	26.4	27.2	30.5	27.0	30.5	26.7	30.1	6.9	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.2	21.6	29.6	26.0	21.4	29.5	25.8	21.1	29.4	86.2	31.0	85.1	30.2	83.7	30.4	30.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.8	10.9	10.3	DB	20.2	32.5	1.3	0.9	19.3	33.2	18.6	33.8	17.9	34.3	16.9	35.0	(4)
(5)			WB	18.5	28.9	1.0	0.7	17.8	29.4	17.2	29.8	16.7	30.2	16.0	30.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.7	26.6	26.5	26.9	27.5	28.3	28.4	28.3	28.5	28.5	28.3	27.8	27.2	(6)	
	DBStd	1.15	0.85	0.90	0.98	0.98	0.76	0.76	0.77	0.87	0.96	0.95	0.90	0.87	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD10.0	6472	514	462	523	524	568	552	569	572	554	567	534	533	(10)	
	CDD18.3	3431	256	228	264	274	310	302	311	314	304	308	284	274	(11)	
	CDH23.3	37513	2335	2014	2493	2842	3575	3532	3640	3783	3694	3637	3162	2806	(12)	
	CDH26.7	11346	381	332	516	730	1168	1188	1230	1404	1444	1323	989	641	(13)	
(14) Wind	WSAvg	6.9	7.6	7.8	7.4	7.2	7.4	8.0	7.5	6.2	5.3	5.6	6.1	7.2	(14)	
(15) Precipitation	PrecAvg	1623	94	57	54	93	94	162	172	176	183	222	198	104	(15)	
	PrecMax	2147	212	177	160	374	246	386	538	296	468	491	409	226	(16)	
	PrecMin	1162	18	0	4	16	25	57	84	77	35	56	31	22	(17)	
	PrecStd	307	44	35	35	82	52	89	88	49	95	99	93	49	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.0	30.1	31.0	31.2	31.9	31.2	31.2	32.1	32.2	32.0	31.2	30.2	(19)	
		MCWB	24.5	24.3	25.2	25.8	26.4	26.4	26.4	26.7	26.8	26.7	26.4	25.3	(20)	
	2%	DB	29.2	29.2	30.1	30.7	31.1	31.1	31.1	31.2	31.8	31.2	30.9	30.1	(21)	
		MCWB	24.2	23.9	24.6	25.5	26.0	26.3	26.4	26.5	26.6	26.5	26.3	25.2	(22)	
	5%	DB	29.0	29.0	29.3	30.1	30.9	30.8	30.8	31.1	31.2	31.1	30.3	29.7	(23)	
		MCWB	24.1	23.8	24.1	25.1	25.8	26.1	26.2	26.5	26.5	26.5	25.9	24.9	(24)	
	10%	DB	28.7	28.6	29.0	29.8	30.2	30.1	30.2	30.8	31.0	30.8	30.1	29.1	(25)	
		MCWB	23.9	23.5	23.8	24.8	25.4	25.8	25.9	26.4	26.5	26.3	25.8	24.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.6	25.5	25.9	26.6	27.2	27.2	27.2	27.5	27.6	27.4	27.2	26.5	(27)	
		MCDB	28.5	28.9	29.5	30.3	30.5	30.3	30.2	30.4	30.7	30.4	30.2	29.3	(28)	
	2%	WB	25.0	24.7	25.2	26.1	26.6	26.7	26.7	27.1	27.1	27.0	26.7	25.9	(29)	
		MCDB	28.3	28.4	29.0	29.6	30.1	29.9	30.0	30.5	30.7	30.6	29.9	28.9	(30)	
	5%	WB	24.6	24.3	24.7	25.6	26.3	26.5	26.5	26.7	26.7	26.7	26.4	25.5	(31)	
		MCDB	27.9	27.9	28.6	29.1	29.7	29.7	29.8	30.1	30.3	30.1	29.6	28.6	(32)	
	10%	WB	24.2	23.9	24.3	25.2	25.9	26.2	26.2	26.5	26.6	26.5	26.1	25.1	(33)	
		MCDB	27.6	27.6	28.1	28.7	29.4	29.3	29.4	29.9	30.1	29.8	29.3	28.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.4	4.5	4.6	4.5	4.3	4.0	4.2	4.7	5.2	5.0	4.8	4.5	(35)	
		MCDBR	4.5	4.7	4.7	4.6	4.6	4.3	4.5	5.0	5.4	5.2	5.0	4.7	(36)	
	5% WB	MCWBR	2.5	2.5	2.4	2.3	2.3	2.3	2.4	2.5	2.6	2.6	2.6	2.5	(37)	
		MCWBR	4.2	4.5	4.5	4.4	4.4	4.2	4.4	4.8	5.2	5.1	4.9	4.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.369	0.385	0.384	0.425	0.469	0.553	0.513	0.456	0.421	0.383	0.379	0.378	(40)	
		taud	2.602	2.533	2.548	2.449	2.305	2.033	2.164	2.386	2.498	2.635	2.630	2.603	(41)	
	Edn at Noon		931	928	931	885	833	757	791	847	884	918	915	913	(42)	
			95	106	106	117	133	172	152	123	110	94	92	93	(43)	
(44) All-Sky Solar Radiation	RadAvg		5.24	5.91	6.33	6.49	6.32	6.02	6.16	6.25	6.04	5.48	5.04	4.97	(44)	
	RadStd		0.18	0.10	0.26	0.26	0.37	0.32	0.26	0.29	0.26	0.23	0.37	0.20	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		+0.27	+0.49	N/A	+0.20	N/A	N/A	N/A	N/A	+113	+112

Nomenclature: See separate page